Lab 5: Normalization – Task 1

Question: List any relations in the Lab 4 Solution that are not in BCNF. Justify.

Answer

None of the relations in the Lab 4 Solution violates BCNF. All 31 relations are in BCNF with respect to the functional dependencies stated in Lab 4.

Rule

A relation R is in BCNF if, for every non-trivial functional dependency X → A that holds on R, X is a superkey of R. To show that a relation is in BCNF, it is enough to check that the determinant (left-hand side) of every non-trivial FD is a superkey.

Relation-by-relation verification

In every relation, the only non-trivial FD listed in Lab 4 has the primary key (or the full composite key) as its determinant. Since that key determines all other attributes, it is a candidate key and therefore a superkey. Relations with no non-key attributes have only trivial FDs, which never violate BCNF.

Relation	Candidate key	Non-trivial FDs	BCNF?
Person	pid	pid → {firstName, lastName, email, affiliation, startDate, endDate}	Yes
Student	pid	pid → program	Yes
Employee	pid	pid → {phone, office, supervisorPid}	Yes
Academic	pid	Only trivial FDs	Yes
NonAcademic	pid	Only trivial FDs	Yes
Faculty	pid	pid → position	Yes
Administrative	pid	pid → position	Yes
Technical	pid	pid → position	Yes
Advises	(academicPid, studentPid)	Only trivial FDs (all-key relation)	Yes
Laboratory	labId	labId → {name, building, roomNumber, discipline, supervisorPid}	Yes
Attached	(pid, labId)	Only trivial FDs (all-key relation)	Yes
ResearchProject	code	code → {title, startDate, endDate, status}	Yes
Participates	(pid, projectCode)	(pid, projectCode) → role	Yes
Budget	budgetLine	budgetLine → {amountGranted, amountDisbursed, startDate, endDate, managerPid}	Yes
FundsPrj	(budgetLine, projectCode)	Only trivial FDs (all-key relation)	Yes
FundsLab	(budgetLine, labId)	Only trivial FDs (all-key relation)	Yes
EquipmentModel	modelId	modelId → {commercialName, manufacturer, category, requiredEnvironment, trainingMandatory}	Yes
EquipmentUnit	serialNo	serialNo → {acquisitionDate, purchaseCost, status, portable, modelId, labId}	Yes
Certification	code	code → {title, issuingAuthority, validityPeriod, safetyLevel}	Yes

Relation	Candidate key	Non-trivial FDs	BCNF?
Requires	(modelId, certCode)	Only trivial FDs (all-key relation)	Yes
Holds	(pid, certCode)	(pid, certCode) → {issueDate, expirationDate, grade}	Yes
Reservation	resId	resId → {submissionTS, plannedStart, plannedEnd, purpose, status, madeByPid, projectCode, approverPid}	Yes
Reserves	(resId, serialNo)	Only trivial FDs (all-key relation)	Yes
Maintenance	(serialNo, startTS)	(serialNo, startTS) → {endTS, type, description, cost, outcome, technicalPid}	Yes
CalibrationRecord	(serialNo, calibDate)	(serialNo, calibDate) → {calibrationType, result, nextDueDate, remarks}	Yes
Consumable	consId	consId → {name, unitOfMeasure, hazardLevel, reorderThreshold}	Yes
Supplier	suppId	suppId → {name, contactEmail, phone}	Yes
Supplies	(suppId, consId, labId)	(suppId, consId, labId) → unitPrice	Yes
Stocks	(labId, consId)	(labId, consId) → {quantityOnHand, lastRestockDate, storageCondition, technicalPid, monitoringSince}	Yes
Consumes	(resId, labId, consId)	(resId, labId, consId) → quantityUsed	Yes

Justification by category

- **Single-attribute key relations** (Person, Student, Employee, Faculty, Administrative, Technical, Laboratory, ResearchProject, Budget, EquipmentModel, EquipmentUnit, Certification, Reservation, Consumable, Supplier): the determinant is the key, so it is a superkey.
- **Composite-key relations with dependent attributes** (Participates, Holds, Maintenance, CalibrationRecord, Supplies, Stocks, Consumes): the determinant is the full composite key, which is a superkey. The Lab 4 solution explicitly notes that no proper subset of the key determines the non-key attributes (e.g. neither pid nor projectCode alone determines role), so there is no partial dependency.
- **All-key relations** (Advises, Attached, FundsPrj, FundsLab, Requires, Reserves) and the single-attribute relations Academic and NonAcademic: there are no non-key attributes, so only trivial FDs hold. BCNF is satisfied automatically.

Conclusion: no relation in the Lab 4 Solution violates BCNF, so no decomposition is required for Task 1.

(a) Equipment Inventory

Given Schema and Functional Dependencies

Schema:

$$\text{EquipmentInventory}(S, M, N, L)$$

where $S =$ serialNo, $M =$ modelId, $N =$ manufacturer, and $L =$ labId. Functional dependencies F:

$$S \to ML, \qquad M \to N.$$

1. Candidate Keys

- **Step 1 (Mandatory attributes):** The attribute S never appears on the RHS of any given FD in F. Therefore, S cannot be derived from any other attributes and must be present in every candidate key.
- **Step 2 (Compute closure S^+):**

$$\begin{aligned} \text{Initialize: } & S^+ = \{S\} \\ \text{Apply } S \to ML: \quad & S^+ = \{S, M, L\} \\ \text{Apply } M \to N: \quad & S^+ = \{S, M, L, N\} = R. \end{aligned}$$

  Since $S^+ = R$, $\{S\}$ is a superkey.
- **Step 3 (Prove minimality):** Because $\{S\}$ is a single-attribute superkey, its only proper subset is $\emptyset$, which cannot determine R. Since every candidate key must contain S, no other minimal key can exist.

Unique Candidate Key: $\{S\}$ (i.e., serialNo)

2. BCNF Evaluation

We check each nontrivial FD in F:

- **FD 1 ($S \to ML$):** The determinant S is the candidate key ($S^+ = R$), so this FD satisfies BCNF.
- **FD 2 ($M \to N$):** Compute the closure of the determinant M:

$$M^+ = \{M, N\} \neq R \quad \text{(missing } S \text{ and } L\text{)}.$$

  Because M is not a superkey and $N \notin \{M\}$, the dependency $M \to N$ violates BCNF.

Satisfies BCNF? No.

3. Violating FD and Strongest Normal Form

- **Violating FD:** $M \to N$ (modelId → manufacturer). Its determinant M is not a superkey because multiple equipment units can share the same modelId, so knowing modelId does not uniquely identify serialNo or labId.
- **3NF Test:** For $M \to N$, M is not a superkey, and the RHS attribute N is *non-prime* (since the only candidate key is $\{S\}$). Thus, **3NF is violated** due to the transitive dependency $S \to M \to N$.

Strongest Normal Form Satisfied: 1NF

4. BCNF Decomposition

Using the violating FD $M \to N$ with closure $M^+ = \{M, N\}$, we split R into:

$$R_1 = M^+ = \{M, N\} \Rightarrow \textbf{ModelInfo}(\underline{\text{modelId}}, \text{manufacturer})$$
$$R_2 = \{M\} \cup (R - M^+) = \{S, M, L\} \Rightarrow \textbf{Equipment}(\underline{\text{serialNo}}, \text{modelId}, \text{labId})$$

BCNF Verification:

- In **ModelInfo**(M, N), $M \to N$ holds with $M^+ = \{M, N\} = R_1$ (and it has only 2 attributes), so it is in BCNF.
- In **Equipment**(S, M, L), the projected nontrivial FD is $S \to ML$. Since $S^+ = \{S, M, L\} = R_2$, S is a key and **Equipment** is in BCNF.

5. Lossless-Join Verification

Compute the intersection of the decomposed schemas:

$$R_1 \cap R_2 = \{M, N\} \cap \{S, M, L\} = \{M\}.$$

Since $M \to N \in F$, we have $M \to MN$, which means $(R_1 \cap R_2) \to R_1 \in F^+$. Because the common attribute modelId is a key of **ModelInfo**, $\pi_{R_1}(r)$ *da* $\pi_{R_2}(r) = r$ holds.

Lossless Join: Yes

6. Dependency Preservation

Projecting F onto R_1 and R_2 yields:

$$F_{R_1} = \{M \to N\}, \qquad F_{R_2} = \{S \to ML\}.$$

Since $F_{R_1} \cup F_{R_2} = \{M \to N, S \to ML\} = F$, we immediately have $(F_{R_1} \cup F_{R_2})^+ = F^+$. Both dependencies can be enforced locally without joins.

Dependency Preservation: Yes

(b) Project Participation

Given Schema and Functional Dependencies

Schema:

$$\text{ProjectParticipation}(P, C, R_o, I)$$

where P = pid, C = projectCode, R_o = role, and I = piPid. Functional dependencies F :

$$PC \rightarrow R_o, \qquad C \rightarrow I.$$

1. Candidate Keys

- **Step 1 (Mandatory attributes):** Neither P nor C appears on the RHS of any FD in F. Thus, every candidate key must contain $\{P, C\}$.
- **Step 2 (Compute closure $(PC)^+$):**

$$\begin{aligned} \text{Initialize:} \quad & (PC)^+ = \{P, C\} \\ \text{Apply } PC \rightarrow R_o: \quad & (PC)^+ = \{P, C, R_o\} \\ \text{Apply } C \rightarrow I: \quad & (PC)^+ = \{P, C, R_o, I\} = R. \end{aligned}$$

  Thus, $\{P, C\}$ is a superkey.
- **Step 3 (Prove minimality):** Check proper subsets of $\{P, C\}$:

$$P^+ = \{P\} \neq R, \qquad C^+ = \{C, I\} \neq R.$$

  Since neither single-attribute subset is a superkey, $\{P, C\}$ is minimal.

Unique Candidate Key: $\{P, C\}$ (i.e., (pid, projectCode))

2. BCNF Evaluation

- **FD 1 ($PC \rightarrow R_o$):** The determinant PC is the candidate key ($(PC)^+ = R$), satisfying BCNF.
- **FD 2 ($C \rightarrow I$):** Compute the closure of C:

$$C^+ = \{C, I\} \neq R \quad \text{(missing } P \text{ and } R_o\text{)}.$$

  Thus, C is not a superkey, and $C \rightarrow I$ violates BCNF.

Satisfies BCNF? No.

3. Violating FD and Strongest Normal Form

- **Violating FD:** $C \rightarrow I$ (projectCode → piPid). Its determinant C is not a superkey because a project can have multiple participants (P), so knowing projectCode alone cannot deter-mine pid or role.
- **3NF Test:** In $C \rightarrow I$, C is not a superkey and I is non-prime ($I \notin \{P, C\}$). Thus, **3NF fails**.

Strongest Normal Form Satisfied: 1NF

4. BCNF Decomposition

Using the violating FD $C \rightarrow I$ with closure $C^+ = \{C, I\}$, we decompose R into:

$R_1 = C^+ = \{C, I\} \Longrightarrow$ **Project**(projectCode, piPid)

$R_2 = \{C\} \cup (R - C^+) = \{P, C, R_o\} \Longrightarrow$ **Participation**(pid, projectCode, role)

BCNF Verification:

- **Project**(C, I) has key C $(C \rightarrow I)$ and only 2 attributes $\Longrightarrow$ in BCNF.
- **Participation**(P, C, R_o) has only one nontrivial FD, $PC \rightarrow R_o$, where $(PC)^+_{R_2} = \{P, C, R_o\} = R_2 \Longrightarrow$ in BCNF.

5. Lossless-Join Verification

The intersection of R_1 and R_2 is:

$$R_1 \cap R_2 = \{C, I\} \cap \{P, C, R_o\} = \{C\}.$$

Since $C \rightarrow I \in F$, we have $C \rightarrow CI = R_1$, meaning $(R_1 \cap R_2) \rightarrow R_1 \in F^+$. Thus, the decomposition is lossless.

Lossless Join: Yes

6. Dependency Preservation

Projecting the dependencies gives:

$$F_{R_1} = \{C \rightarrow I\}, \qquad F_{R_2} = \{PC \rightarrow R_o\}.$$

Since $(F_{R_1} \cup F_{R_2})^+ = \{C \rightarrow I, PC \rightarrow R_o\}^+ = F^+$, all functional dependencies are preserved locally.

Dependency Preservation: Yes

(c) Equipment Certification Requirement

Given Schema and Functional Dependencies

Schema:

$$\text{EquipmentCertification}(M, C, T, A)$$

where $M =$ modelId, $C =$ certCode, $T =$ certTitle, and $A =$ issuingAuthority.

Functional dependencies F:

$$C \rightarrow TA$$

Note: Because the problem statement specifies that (modelId, certCode) identifies an equipment-model certification requirement (and many-to-many associations hold between M and C), $\{M, C\}$ is the key of the relation while $C \rightarrow TA$ is the specific dependency holding on certification codes.

1. Candidate Keys

- **Step 1 (Mandatory attributes):** Neither M nor C appears on the RHS of any nontrivial FD. Thus, every candidate key must contain $\{M, C\}$.
- **Step 2 (Compute closure $(MC)^+$):**

$$(MC)^+ = \{M, C\} \xrightarrow{C \rightarrow TA} \{M, C, T, A\} = R.$$

  Thus, $\{M, C\}$ is a superkey.
- **Step 3 (Prove minimality):** Check single-attribute subsets:

$$M^+ = \{M\} \not\models R, \qquad C^+ = \{C, T, A\} \not\models R \quad (\text{since } M \notin C^+).$$

  Neither proper subset is a superkey, so $\{M, C\}$ is minimal.

Unique Candidate Key: $\{M, C\}$ (i.e., (modelId, certCode))

2. BCNF Evaluation

Examine the nontrivial FD $C \rightarrow TA$:

$$C^+ = \{C, T, A\} \not\models R \quad (\text{missing } M).$$

Since C is not a superkey of R, $C \rightarrow TA$ violates BCNF.

Satisfies BCNF? No.

3. Violating FD and Strongest Normal Form

- **Violating FD:** $C \rightarrow TA$ (certCode → {certTitle, issuingAuthority}). Its determinant C is not a superkey because a single certification may be required by several equipment mod-els (M).
- **3NF Test:** Neither T nor A is a prime attribute (only M and C are prime), and C is not a superkey $\Longrightarrow$ **3NF fails**.

Strongest Normal Form Satisfied: 1NF

4. BCNF Decomposition

Using the violating FD $C \rightarrow TA$ with closure $C^+ = \{C, T, A\}$, we decompose R into:

$R_1 = C^+ = \{C, T, A\} \Longrightarrow$ **Certification**(<u>certCode</u>, certTitle, issuingAuthority)
$R_2 = \{C\} \cup (R - C^+) = \{M, C\} \Longrightarrow$ **ModelCertification**(<u>modelId, certCode</u>)

BCNF Verification:

- In **Certification**(C, T, A), the only nontrivial FD is $C \rightarrow TA$, and $C^+_{R_1} = \{C, T, A\} = R_1$, so C is a superkey $\Longrightarrow$ in BCNF.
- In **ModelCertification**(M, C), there are only two attributes (and no nontrivial FDs $M \rightarrow C$ or $C \rightarrow M$ hold), so the composite pair $\{M, C\}$ is the key $\Longrightarrow$ automatically in BCNF.

5. Lossless-Join Verification

The common attribute set is:

$$R_1 \cap R_2 = \{C, T, A\} \cap \{M, C\} = \{C\}.$$

Since $C \rightarrow TA \in F$, we have $C \rightarrow CTA = R_1$, i.e., $(R_1 \cap R_2) \rightarrow R_1 \in F^+$. The join is lossless.

Lossless Join: Yes

6. Dependency Preservation

The only non-trivial FD C → TA is projected directly onto R1 (F_{R1} = {C → TA}). R2 = ModelCertification(M, C) has no non-trivial FDs ($F_{R2} = \emptyset$); its key {M, C} is enforced as its primary key.

Thus, $(F_{R_1} \cup F_{R_2})^+ = F^+$.

Dependency Preservation: Yes

(d) Supplier Offer

Given Schema and Functional Dependencies

Schema:

$$\text{SupplierOffer}(S, C, L, N, U)$$

where $S =$ suppId, $C =$ consId, $L =$ labId, $N =$ supplierName, and $U =$ unitPrice. Functional dependencies F:

$$S \to N, \qquad SCL \to U.$$

1. Candidate Keys

- **Step 1 (Mandatory attributes):** The attributes S, C, and L never appear on the RHS of any FD in F. Consequently, all three must belong to every candidate key: $\{S, C, L\}$.
- **Step 2 (Compute closure $(SCL)^+$):**

$$\begin{aligned} \text{Initialize:} \quad & (SCL)^+ = \{S, C, L\} \\ \text{Apply } S \to N: \quad & (SCL)^+ = \{S, C, L, N\} \\ \text{Apply } SCL \to U: \quad & (SCL)^+ = \{S, C, L, N, U\} = R. \end{aligned}$$

  Thus, $\{S, C, L\}$ is a superkey.
- **Step 3 (Prove minimality):** Since S, C, and L are all mandatory (none appears on any RHS), dropping any one of them makes it impossible to derive the dropped attribute. Specifically:

$$(SC)^+ = \{S, C, N\} \not\models R, \quad (SL)^+ = \{S, L, N\} \not\models R, \quad (CL)^+ = \{C, L\} \not\models R.$$

  Hence, $\{S, C, L\}$ is minimal.

Unique Candidate Key: $\{S, C, L\}$ (i.e., (suppId, consId, labId))

2. BCNF Evaluation

- **FD 1 ($SCL \to U$):** The determinant SCL is the candidate key ($(SCL)^+ = R$), satisfying BCNF.
- **FD 2 ($S \to N$):** Compute the closure of S:

$$S^+ = \{S, N\} \not\models R \quad \text{(missing } C, L, U\text{)}.$$

  Since S is not a superkey, $S \to N$ violates BCNF.

Satisfies BCNF? No.

3. Violating FD and Strongest Normal Form

- **Violating FD:** $S \to N$ (suppId → supplierName). Its determinant S is not a superkey because a supplier can make offers for multiple consumables (C) and laboratories (L).
- **3NF Test:** In $S \to N$, S is not a superkey and N is non-prime ($N \notin \{S, C, L\}$) $\Longrightarrow$ **3NF fails**.

Strongest Normal Form Satisfied: 1NF

4. BCNF Decomposition

Using the violating FD $S \rightarrow N$ with closure $S^+ = \{S, N\}$, we decompose R into:

$$R_1 = S^+ = \{S, N\} \Longrightarrow \textbf{Supplier}(\underline{\text{suppId}}, \text{supplierName})$$

$$R_2 = \{S\} \cup (R - S^+) = \{S, C, L, U\} \Longrightarrow \textbf{Offer}(\underline{\text{suppId, consId, labId}}, \text{unitPrice})$$

BCNF Verification:

- **Supplier**(S, N) has key S ($S \rightarrow N$) and 2 attributes $\Longrightarrow$ in BCNF.
- **Offer**(S, C, L, U) has only the nontrivial FD $SCL \rightarrow U$, where $(SCL)^+_{R_2} = \{S, C, L, U\} = R_2 \Longrightarrow$ in BCNF.

5. Lossless-Join Verification

The common attribute set is:

$$R_1 \cap R_2 = \{S, N\} \cap \{S, C, L, U\} = \{S\}.$$

Since $S \rightarrow N \in F$, we have $S \rightarrow SN = R_1$, so $(R_1 \cap R_2) \rightarrow R_1 \in F^+$. The decomposition is lossless.

Lossless Join: Yes

6. Dependency Preservation

Projecting F onto R_1 and R_2 gives:

$$F_{R_1} = \{S \rightarrow N\}, \qquad F_{R_2} = \{SCL \rightarrow U\}.$$

Since $F_{R_1} \cup F_{R_2} = F$, we have $(F_{R_1} \cup F_R)^+_2 = F^+$. Both dependencies are preserved locally without joins.

Dependency Preservation: Yes

(e) Reservation and Project Information

Given Schema and Functional Dependencies

Schema:

$$\text{ReservationProject}(I, C, T, P, S)$$

where $I =$ resId, $C =$ projectCode, $T =$ projectTitle, $P =$ madeByPid, and $S =$ status. Functional dependencies F :

$$I \to CPS, \qquad C \to T.$$

1. Candidate Keys

- **Step 1 (Mandatory attributes):** The attribute I never appears on the RHS of any FD in F. Thus, I must belong to every candidate key.
- **Step 2 (Compute closure I^+):**

$$\begin{aligned} \text{Initialize:} \quad & I^+ = \{I\} \\ \text{Apply } I \to CPS: \quad & I^+ = \{I, C, P, S\} \\ \text{Apply } C \to T: \quad & I^+ = \{I, C, P, S, T\} = R. \end{aligned}$$

  Thus, $\{I\}$ is a superkey.
- **Step 3 (Prove minimality):** Since $\{I\}$ is a single-attribute superkey, its only proper subset is $\emptyset$. Moreover, because every candidate key must contain I, no other minimal key exists.

Unique Candidate Key: $\{I\}$ (i.e., resId)

2. BCNF Evaluation

- **FD 1 ($I \to CPS$):** The determinant I is the candidate key ($I^+ = R$), satisfying BCNF.
- **FD 2 ($C \to T$):** Compute the closure of C:

$$C^+ = \{C, T\} \neq R \quad \text{(missing } I, P, S\text{)}.$$

  Since C is not a superkey, $C \to T$ violates BCNF.

Satisfies BCNF? No.

3. Violating FD and Strongest Normal Form

- **Violating FD:** $C \to T$ (projectCode → projectTitle). Its determinant C is not a superkey because multiple reservations (I) can be made under the same project (C).
- **3NF Test:** In $C \to T$, C is not a superkey and T is non-prime ($T \notin \{I\}$). Thus, **3NF fails** due to the transitive dependency $I \to C \to T$.

Strongest Normal Form Satisfied: 1NF

4. BCNF Decomposition

Using the violating FD $C \rightarrow T$ with closure $C^+ = \{C, T\}$, we decompose R into:

$R_1 = C^+ = \{C, T\} \Longrightarrow$ **Project**(projectCode, projectTitle)

$R_2 = \{C\} \cup (R - C^+) = \{I, C, P, S\} \Longrightarrow$ **Reservation**(resId, projectCode, madeByPid, status)

BCNF Verification:

- **Project**(C, T) has key C ($C \rightarrow T$) and 2 attributes $\Longrightarrow$ in BCNF.
- **Reservation**(I, C, P, S) has projected nontrivial FD $I \rightarrow CPS$, where $I^+_{R_2} = \{I, C, P, S\} = R_2 \Longrightarrow$ in BCNF.

5. Lossless-Join Verification

The common attribute is:

$$R_1 \cap R_2 = \{C, T\} \cap \{I, C, P, S\} = \{C\}.$$

Since $C \rightarrow T \in F$, we have $C \rightarrow CT = R_1$, meaning $(R_1 \cap R_2) \rightarrow R_1 \in F^+$. The decomposition is lossless.

Lossless Join: Yes

6. Dependency Preservation

Projecting F onto R_1 and R_2 gives:

$$F_{R_1} = \{C \rightarrow T\}, \qquad F_{R_2} = \{I \rightarrow CPS\}.$$

Since $F_{R_1} \cup F_{R_2} = F$, we have $(F_{R_1} \cup F_{R_2})^+ = F^+$ (including the transitive dependency $I \rightarrow T$ via $I \rightarrow C$ and $C \rightarrow T$).

Dependency Preservation: Yes

Final Summary Table of All Five Cases

Case	Candidate Key	BCNF?	Strongest NF	Violating FD	BCNF Decomposition (Lossless & Preserving)
(a)	{serialNo}	No	**1NF** (not in 3NF)	$M \rightarrow N$	**ModelInfo**(<u>*M*</u>, *N*) **Equipment**(<u>*S*</u>, *M*, *L*)
(b)	{pid, projectCode}	No	**1NF** (not in 3NF)	$C \rightarrow I$	**Project**(<u>*C*</u>, *I*) **Participation**(<u>*P*, *C*</u>, R_o)
(c)	{modelId, certCode}	No	**1NF** (not in 3NF)	$C \rightarrow TA$	**Certification**(<u>*C*</u>, *T*, *A*) **ModelCertification**(<u>*M*, *C*</u>)
(d)	{suppId, consId, labId}	No	**1NF** (not in 3NF)	$S \rightarrow N$	**Supplier**(<u>*S*</u>, *N*) **Offer**(<u>*S*, *C*, *L*</u>, *U*)
(e)	{resId}	No	**1NF** (not in 3NF)	$C \rightarrow T$	**Project**(<u>*C*</u>, *T*) **Reservation**(<u>*I*, *C*, *P*, *S*</u>)

Global Conclusion: For all five relations **(a)–(e)**, the top-down BCNF decomposition achieves **Lossless Join = Yes** and **Dependency Preservation = Yes**.